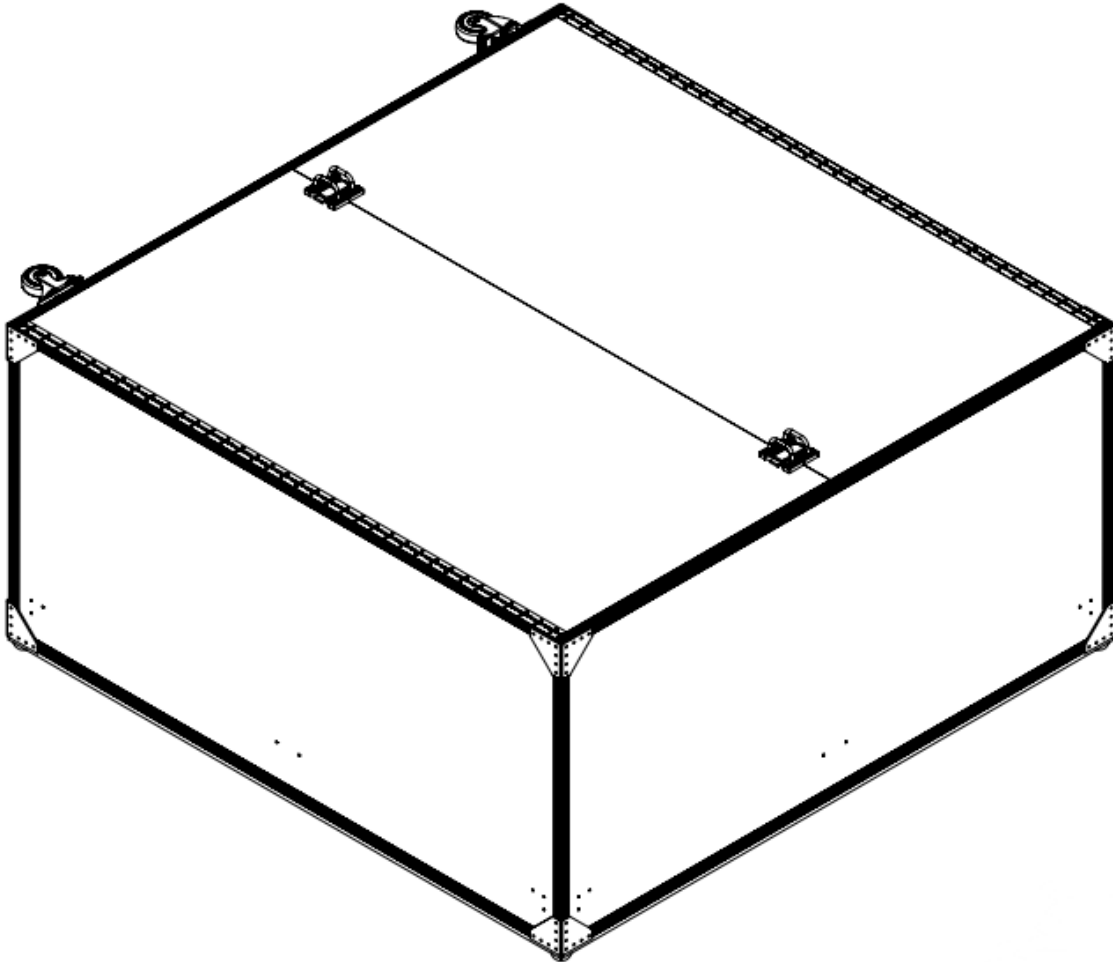


Fall 2023 Robotics Competition Arena Proposal

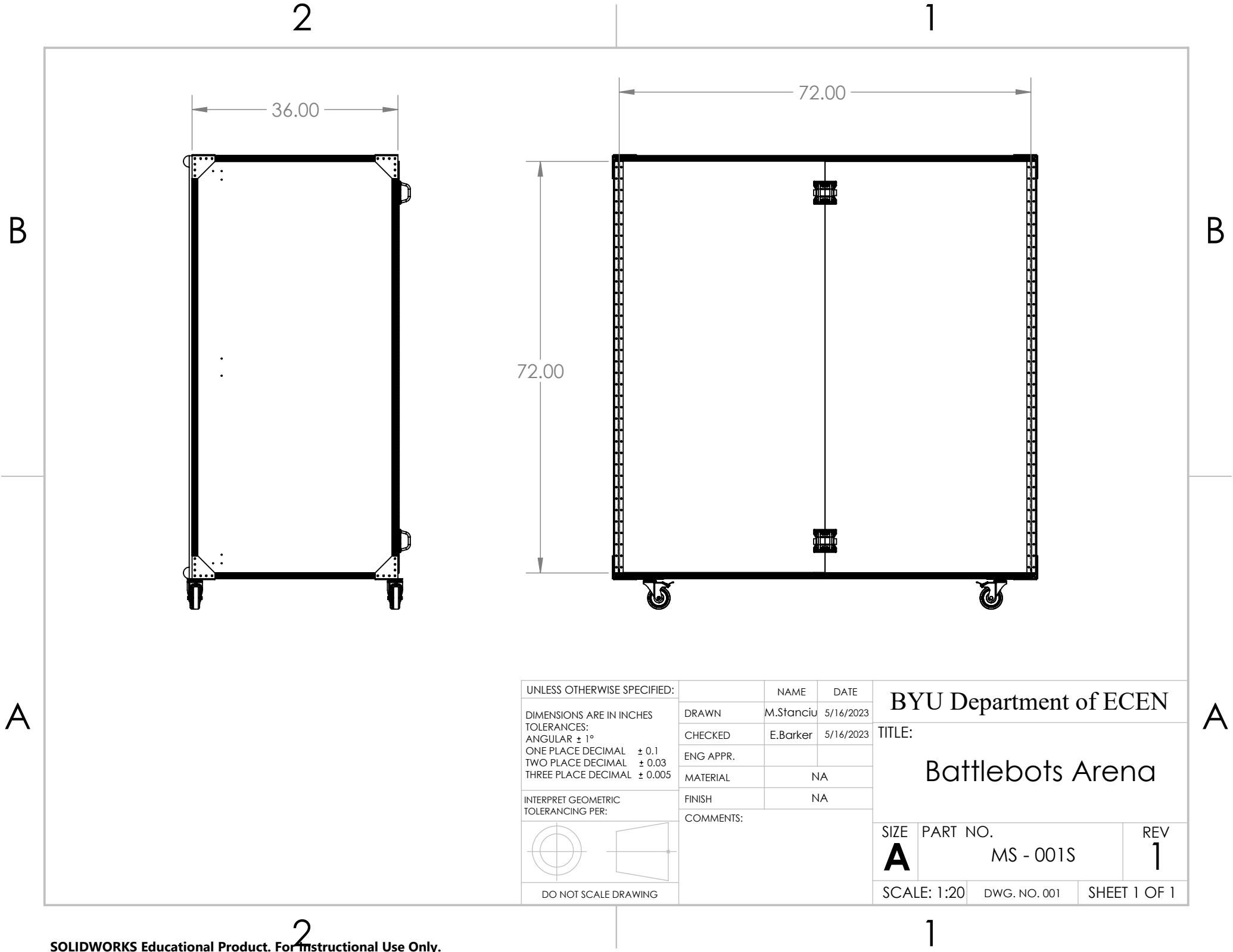


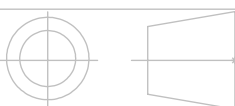
Mihai Stanciu

BYU Department of Electrical and Computer Engineering

Experiential Learning Center

June, 2023



UNLESS OTHERWISE SPECIFIED:		NAME	DATE	BYU Department of ECEN			
DIMENSIONS ARE IN INCHES TOLERANCES: ANGULAR ± 1° ONE PLACE DECIMAL ± 0.1 TWO PLACE DECIMAL ± 0.03 THREE PLACE DECIMAL ± 0.005 INTERPRET GEOMETRIC TOLERANCING PER: 		DRAWN	M.Stanciu	5/16/2023	TITLE: Battlebots Arena		
		CHECKED	E.Barker	5/16/2023			
		ENG APPR.					
		MATERIAL	NA				
		FINISH	NA				
		COMMENTS:			SIZE	PART NO.	REV
					A	MS - 001S	1
DO NOT SCALE DRAWING					SCALE: 1:20	DWG. NO. 001	SHEET 1 OF 1

2

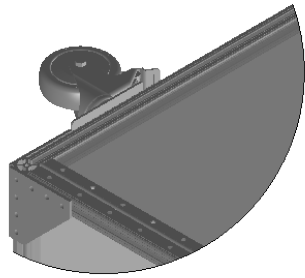
1

B

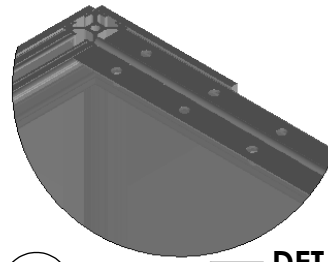
B

A

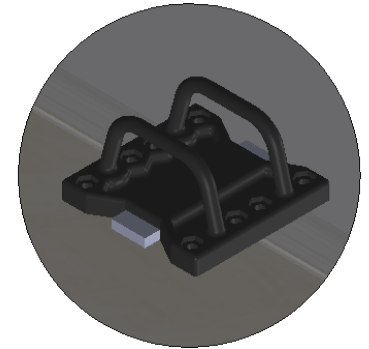
A



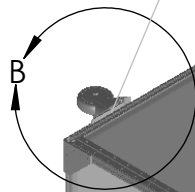
DETAIL B
Casters for Easier Transportation



DETAIL A
Piano Hinge for Trapdoor Stability



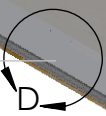
DETAIL C
Locking Bar Mechanism
to Ensure that
Trapdoors Remain Closed



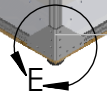
B



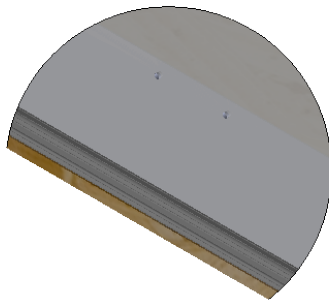
C



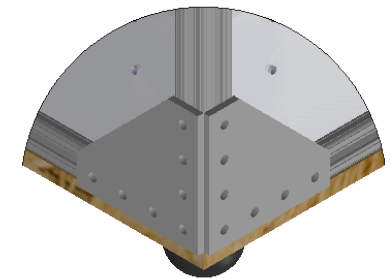
D



E



DETAIL D
Double Polycarbonate Sheets
and Aluminum Plate
to Minimize Potential Wall Damage



DETAIL E
Aluminum 8020 Strut and Right
Angle Brackets for Increased
Stability

Arena Design Features

2

1

Bill of Materials per Unit

QTY	Part	Distributor	Cost	Total Cost	Product Link
3	Lexan Pannel (6 Pack)	Home Depot	\$650	\$1,950	Home Depot
8	8020 Strut (6 ft)	McMaster	\$34.03	\$272	McMaster Carr
8	8020 Strut (3 ft)	McMaster	\$16.65	\$133	McMaster Carr
16	Right Angle Bracket	McMaster	\$21.01	\$336	McMaster Carr
4	Aluminum Plate	Shop Provided	N/A	N/A	N/A
2	OBS Board (4x8 ft)	Home Depot	\$20.42	\$41	Home Depot
4	Caster	McMaster	\$66.21	\$265	McMaster Carr
4	Rubber Foot	McMaster	\$9.09	\$36	McMaster Carr
2	Piano Hinge	McMaster	\$11.37	\$23	McMaster Carr
4	8020 Screw/Nut (40 Pcs Set)	Amazon	\$18.99	\$76	Amazon
4	1/4-20 3/4" Screw (25 Pcs Set)	Amazon	\$9.49	\$38	Amazon
1	1/4-20 Nut (100 Pcs Set)	Amazon	\$6.73	\$7	Amazon
4	PLA Handle	3D Print	\$4.5	\$18	N/A
2	PLA Support Bracket	3D Print	\$3	\$6	N/A
2	Steel Bar	Shop Provided	N/A	N/A	N/A
				Total Cost per Unit:	\$3,201